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Summary

Omar Lizardo's work provides a comprehensive theoretical framework for the study of **culture and networks**, arguing that this diverse field is anchored by two primary conceptual "ur-models." The first, **constructuralism**, explores the dynamic feedback loop where the exchange of cultural information fosters social ties, which in turn facilitate further cultural similarity and the creation of **cultural niches**. The second framework, the **cultural holes imagery**, focuses on the gaps within **multimode networks** and how bridging these absences of relation can drive **creativity, innovation, and social status**. By synthesizing various empirical studies into these two categories, Lizardo shifts the scholarly focus from topical intersections to the **underlying theoretical foundations** that define how social structures and cultural meanings mutually constitute one another. Ultimately, the text serves to organize the "cluttered landscape" of contemporary research into a **coherent scholarly domain** based on the dual processes of cultural conversion and positional brokerage.

Keywords: Culture network theory, Constructural imagery, Cultural holes model, Culture conversion mechanism, Associative diffusion model

Content

13 IntroductIon **There is something paradoxical about 'culture networks' as a field of study. On the one hand, the area is coherent and visible enough to have been the subject of numerous chapter-length reviews in the last decade or so (Breiger & Puetz, 2015; Mützel & Breiger, 2020; Pachucki & Breiger, 2010; Rule & Bearman, 2015), including pro-grammatic treatments by Paul DiMaggio (2011) and Ann Mische (2011) in the previous edition of this handbook.1 However, it is still being deter-mined what exactly the field of culture and net-work studies is as a scholarly domain of practice. Some (Breiger & Puetz, 2015, p. 557) refer to it 'as a research specialty in its own right' – a some-what optimistic assessment – but others see it, at**

least implicitly, more as a cluster of studies united by thematic similarities and substantive emphases across various distinct sub-areas in sociology and the social sciences more generally (DiMaggio, 2011; Mische, 2011; Rule & Bearman, 2015). These include, inter alia, studies of artistic, col-laboration, civil society and creative networks, or the formal modelling of certain cultural elements, such as text, practices, language, concepts, narra-tives and events, as a network of elements, or the study of the meaningful bases of interaction and relationship formation in social networks via cul-tural change and the construction of identities and boundaries. Nevertheless, something is missing in these previous treatments: a general sense of whether, beyond topical intersections of culture and networks or the study of interesting empirical phenomena, there exists a general theoretical or conceptual basis uniting the field. In this chapter, I argue that the answer to this last question is affirmative. I broadly follow Borgatti & Lopez-Kidwell's (2011) approach – see also Borgatti & Halgin (2011) – in the previous edition of this volume to define the parameters of network theory in the case of the study of culture and net-works. I will not review specific pieces of empirical research clustered according to thematic criteria loosely concerned with the culture-networks link-age. Instead, I organised particular strands of the literature according to underlying theoretical ur-models, explicitly stating their assumptions and critical empirical implications and showing how analysts deploy them for analytic and explanatory purposes. Thus, the chapter's primary goal is to outline the theoretical foundations of work link-ing culture and networks. As such, my review of specific research will be highly selective; the point is not to exhaustively consider the large number of studies that have been conducted across the the-matic areas mentioned earlier. Instead, I will select Culture and Networks O m a r L i z a r d o Culture and networks 189 specific exemplars (in Kuhn's sense) in which the core theoretical imagery is showcased or signifi-cantly developed. **The central claim is that just like the seemingly cluttered landscape of network theorising turns out to be organised by a surprisingly small number of underlying theoretical models – only two in Borgatti & Lopez-Kidwell's (2011) accounting – much of the work on the culture-networks linkage is also organised by a surprisingly compact set of theo-retical images. Only two in my accounting. First, a vision of the interplay between personal culture and network ties built on the constructural imagery (Carley, 1991), in which the continuous exchange of cultural information helps people form, construct and maintain social connections while defining sociocultural affinities and boundaries. Second, a vision of the link between people and broader social categories, public culture, groups, cultural objects, genres,**

knowledge and the like, animated by the cultural holes imagery (Pachucki & Breiger, 2010). Here linkages between people and other social enti-ties in multimode networks define distributions of positions – for people and objects – in dually con-stituted 'sociocultural' and 'culturosocial spaces' (Lee & Martin, 2018; Puetz, 2017). Together, the constructural and cultural holes imageries, as well as their recent developments and refinements, form the theoretical backbone of research in culture and networks. The rest of the chapter is organised into two major sections, covering the constructural and cul-tural holes models, their main variations and core applications. I close with a brief section discussing the argument's implications for future work. the constructural Imagery Constructuralism and Network Ecologies **Carley's Constructural Model I begin by considering what, to my knowledge, was the first formal theory of the dynamic cou-pling of culture and networks: Kathleen Carley's (1991) constructural model, one of the ur-models from which a whole swath of research on culture and networks derives. Because it was initially stated as a formal computational model, we do not need to work too hard to uncover the fundamental premises of the model. In the constructural imagery, agents exchange "pieces" of culture (e.g., facts, information, beliefs, tastes, preferences and the like) when they interact. As a result of** interacting, agents become more culturally similar. Moreover, agents' cultural similarity increases the chance of future interaction as the probability of two actors interacting at any given occasion is a positive function of their shared culture. There is independent theoretical reasoning and quite a lot of empirical work that supports each constructural model's premises (considering this work, however, is outside the scope of this chapter). The key idea is that the three premises define a positive feedback loop connecting cul-tural exchange to the 'tie strength' between people (Granovetter, 1973; Marsden & Campbell, 1984), which becomes indexed to their relative cultural similarity compared to that they have with other actors in the system. The more people interact, the more culture they exchange, and the more their dyadic ties strengthen, which leads to future bouts of interaction and dyadic cultural exchange. As Carley (1991) noted in the initial statement of the theory, a plausible equilibrium, all else equal, is for all actors to reach maximum cultural similarity with one another and thus form a fully connected clique in terms of their interaction probabilities. Of course, complications can be introduced to the model to account for the apparent fact of cultural differentiation and differential probabilities of interaction in large-scale systems, inclusive of the capacity for actors to 'lose' (e.g., forget) cultural knowledge (Mark, 1998a) and the emergence of non-interactive 'actants' (books or other informa-tion archives such as the World Wide Web) from which individuals can extract cultural knowledge not available via face-to-face interaction with others (Carley, 1995). Carley's basic construc-tural imagery thus provides the basis for several more recent developments in theorising the link between culture and social networks and linking theory to empirical evidence. Mark's Cultural Ecology Model The first model of the culture and networks linkage explicitly built on the constructural imagery is Mark's (2003) network ecology model (NEM) – see also Mark (1998b) for a precursor. Mark departs from the evident observation that cultural tastes, opinions and beliefs are segregated into distinct 'niches' of socio-demographic space. Here Mark draws on McPherson's (2004) socio-demographic niche theory, which builds on earlier insights from Blau's (1977) macrostructural theory conceptualis-ing social space as a multidimensional space. People's position in this 'Blau space', whose axes are composed of 'dimensions of association', con-tinuous, ordinal and nominal (such as age, gender, race, occupational prestige and the like), shaping the chances people will form network connections. The SAGe hAndbook of SociAl neTwork AnAlySiS190 The basic principle here is that of homophily (McPherson et al., 2001), such that the probability of two being tied in a social network is a function of the number of socio-demographic characteristics they share translates to a distance gradient in Blau space. People close to one another are more likely to be connected. In the NEM, connectivity is a conduit via which cultural tastes, opinions and beliefs are transmit-ted via imitation and social influence. The sug-gested mechanism is that acquiring cultural tastes is best envisioned as a 'local bandwagon' process of network-based imitation or contagion. Because the distribution of network ties in Blau space is 'lumpy' (with people concentrated along areas that maxim-ise similarity), so will the distribution of tastes, which will concentrate along particular 'niches' (particular areas of Blau space). For instance, rock and roll will be more likely to be found in the Blau space area marked by relatively high education, occupational prestige, older age and white ethno-racial identity. Mark's NEM uses the constructural imagery to provide a formal model of how this process of concentration of tastes along socio-demographic can be realised via routine inter-action, thus explaining why tastes and opinions cluster in specific 'cultural niches' and why at any one moment, we are liable to find systematic cor-relations between socio-demographic position and cultural tastes (Bourdieu, [1979]1984). More recent work by DellaPosta et al. (2015) uses insights from the NEM to explain the associa-tion between cultural tastes and practices and pre-sumably (e.g., logically) unrelated attitudes and beliefs such as political liberalism and conserva-tism. The basic idea is that once relatively small correlations between politics and lifestyle begin to obtain (even if for contingent reasons), and politi-cal ideology becomes a fundamental dimension of association in Blau's sense, arbitrary linkages between politics, tastes and lifestyle practices can be amplified via the twin mechanisms of homo-phily and personal influence. Accordingly, local bandwagon processes segregate conservatives and liberals in distinct cultural niches, reifying politi-cal boundaries by the standard mechanisms of relational segregation and selective interaction. Conversion and Matching Lizardo's Culture Conversion Model If

Mark’s cultural ecology model theorised how social-structural biases led to the social segre-gation of cultural tastes (thus relying on half of the two-sided dynamic process of conversion **emphasised by Carley), Lizardo’s (2006) culture conversion model (CCM) examined the obverse process; how cultural tastes can lead to the biased formation of social network ties. Lizardo’s CCM synthesised three key theoretical strands in the literature that, at the time, had developed sepa-rately, despite each addressing critical processes and mechanisms linking culture and social net-works. The first was, of course, Carley’s construc-tural theory (Carley, 1991). The second consisted of micro-interactionist approaches to theorising the functions of culture consumption in modern artistic classification systems linking cultural con-sumption to Simmelian sociability and the crea-tion of bounded solidarities via the mobilisation of cultural capital in ‘interaction rituals’ (DiMaggio, 1987). The third was Bourdieu’s (1986) imagery of the interconvertibility of the three forms of capital (social, economic and cultural). The basic idea is that cultural resources, particularly the pos-session of embodied abilities to consume certain forms of culture, should lead people to differen-tially cumulate and maintain social network ties (such that people with a wide variety of tastes should maintain more extensive social networks). The CCM thus stands in contrast to models postu-lating a one-directional arrow of causation (or conversion) going exclusively from network ties to cultural tastes, like Mark’s NEM. Moreover, according to the CCM, consumption of ‘asset-specific’ cultural goods (e.g., requiring esoteric or difficult-to-acquire cultural knowledge) should have restricted conversion value, lead-ing mainly to creating and maintaining networks of strong ties and local solidarities. On the other hand, consumption of cultural goods that are less asset-specific, such as popular culture with which most people are familiar, should have generalised conversion value, leading to the formation and maintenance of ego networks rich in weak ties. ‘Omnivorous’ consumption of both types of cul-ture should lead to more extensive networks con-taining weak and strong ties. Using data from the culture and network modules of the US General Social Survey (GSS), Lizardo found strong sup-port for the general outlines of the CCM. The more cultural activities people consume, the more exten-sive their reported ego networks are. Highbrow culture – a type of asset-specific culture – increased the volume of strong ties, while consumption of popular culture increased the volume of weak ties. Omnivores who engage in both forms of culture can thus wield complementary resources, enjoying the advantages of bridging (with their weak ties) and bonding (with their strong ties).** Subsequent work has elaborated Lizardo’s culture conversion model conceptually while providing partial empirical support for some of its Culture and networks 191 **key predictions. For example, Schultz & Breiger (2010) rework Lizardo's original distinction between 'asset-specific' and 'non-asset-specific' culture to align with Granovetter's (1973) clas-sic distinction between weak and strong ties. According to Schultz and Breiger, popular cul-ture endowed with generalised conversion can best be considered weak culture. Like weak ties in Granovetter's theory, weak culture turns out to be strong because it allows people to form social ties, however fleeting, linking to others in distant positions in social space. In addition, they show, using US GSS data, that the more mild positive cultural preferences a people have (e.g., 'likes' instead of 'likes very much'), the more likely it is that they perceive the US to be 'united' (rather than divided). In short, weak culture leads to the perception of a potential for interaction across critical social divides, an unexpected implication of the culture conversion imagery, with critical implications for contemporary issues like cultural and political polarisation (DellaPosta et al., 2015). Vaisey and Lizardo's Cultural Matching Model Vaisey & Lizardo's (2010) cultural matching model (CMM) also builds on the constructural imagery linking culture to social network ties. According to the CMM, cultural tastes, values and preferences affect social networks mainly by serv-ing as the underlying basis for homophily (thus reversing the causal arrow postulated by Mark's NEM). In this way, tastes, and other internalised cultural aptitudes, preferences and practices can have an independent causal effect in shaping social networks because people use the match between their tastes and others to self-select into particular social ties (Shalizi & Thomas, 2011). Moreover, the degree of cultural match between two people also determines whether certain social relationships stick over time or instead selectively die off, thus linking cultural matching at the dyadic level with processes of tie-decay at the network level (Burt, 2000; Martin & Yeung, 2006). Using longitudinal data from the US National Study of Youth and Religion, Vaisey and Lizardo find that, compared to those who express more communitarian cultural worldviews, adoles-cents abiding by individualist-expressivist cultural worldviews are more likely to maintain social ties with other adolescents who engage in substance use. In contrast, those abiding by a more individualist-utilitarian worldview are more likely to keep social ties with those who volunteer.** Other empirical work supports critical pre-dictions of the CMM. For instance, Friemel (2012/17) shows that shared cultural tastes **promote tie-formation among adolescents, with similar results using college-aged populations in the domain of musical taste reported by Lomi and Stadtfeld (2014), Vlegels (2014, pp. 71ff) and Hachen et al. (2022). Edelmann and Vaisey (2014), using data from the Cambridge College Network Dataset, a longitudinal sample of graduate**

stu-dents in England, extend the CMM to consider not just substantive matches in terms of tastes and worldviews but also matching in terms of absten-tions or dislikes. Just like people can match their avowed cultural tastes and practices, they can match according to what they do not or refuse to do; refusals are as important as choices (Bourdieu, [1979]1984). Consistent with the extended CMM, Edelmann & Vaisey find that mutual consumption of the same musical genres and common non-consumption systematically affects network ties, increasing the odds that two students will sustain a network connection over time. Lewis and Kaufman's Generalised Conversion Model The most sustained elaboration of the conversion and matching models, theoretically and empirically, is that developed by Lewis and Kaufman (2018). Lewis and Kaufman synthesise the conversion and matching ideas into a generalised conversion model (GCM) while also considering variations in the local cultural ecology to specify the notions of weak and strong culture. Lewis and Kaufman's GCM offers various conceptual advances over previous formulations of the link between culture and net-works based on the constructural imagery. First, they endogenise Schultz and Breiger's idea of weak and strong culture to what they refer to as the local cultural ecology. Rather than using exogenous crite-ria to determine what counts as weak or strong culture (e.g., broad labels such as highbrow or popular), they note that what counts as weak or strong culture will depend on the distribution of tastes and aptitudes in the local cultural environ-ment. Second, they distinguish between different culture conversion and matching mechanisms and show how they can operate in tandem. **First, there is the dyadic conversion of cultural into social capital, whereby people exploit com-monalities in cultural tastes and aptitudes specific to the focal dyad to form and sustain relationships. Second, there is a generalised conversion process, whereby particular forms of taste and cultural con-sumption lead people to form more ties with others (increasing the acquaintance volume). Finally, there is cultural matching, where similarities in entire cul-tural profiles – including, following Edelmann and Vaisey, both active engagements and abstentions – increase the chances of people creating social** The SAGe hAndbook of SociAl neTwork AnAlySiS192 connections with similar others. Using stochastic actor-based models for longitudinal network data on a unique Facebook dataset – the 'tastes, ties and time' data collected during the platform's early days in the US (Lewis et al., 2008) – Lewis & Kaufmann find support for all three conversion mechanisms. Notably, the more tastes two students shared, the more likely they were to become Facebook friends, especially if those tastes were 'specialised' to the local cultural ecology. In the same way, individuals who displayed typical tastes in the local cultural ecology were more likely to accumulate a larger volume of acquaintances via the generalised conversion mechanism. **Constructuralism 2.0: associative diffusion In a recent paper, Goldberg and**

Stein (2018) pro-pose an associative diffusion model (ADM) of the way culture and networks interact. The ADM's argument is deceptively simple but ultimately far-reaching. According to Goldberg and Stein, culture does not diffuse like a 'virus' in social networks; people do not pick up single beliefs, attitudes, tastes, or styles from their contacts. Instead, cul-tural practices, beliefs and attitudes are embedded in a relational network of meanings. That is, there is a 'cultural network' that dictates 'what goes with what' that is analytically independent of the net-work dictating who is tied to whom (a key point of difference with the constructural imagery). In contrast to constructuralism, the ADM uses recent advances in conceptualising culture as schemas (Goldberg, 2011) and schemas as an analytically distinct network of cognitive associations between cultural items that can differ across people. Thus, there are as many cultural networks as there are agents, and what diffuses between people when they interact are the relations between elements. In this way, the ADM can be considered the construc-tural model 2.0. Accordingly, we must distinguish the diffusion of cognitive associations between elements (this is what Goldberg and Stein refer to as 'associative diffusion') or the cognitive representation of 'what goes with what' each agent carries with them from the specific attitudes (e.g., like/dislike, adoption/ refusal, endorsement/non-endorsement) specific people take towards the given practice or attitude. If we allow for associations between elements to be the unit of diffusion, then specific attitudes of adoption/non-adoption towards those elements can be derived as a process of soft-constraint optimi-sation on the part of people (formalising the idea of cognitive consistency between preferences), allowing them to harmonise their attitudes towards particular objects, with the way they believe those cultural elements are linked. For instance, a person may know that caviar is connected to champagne, which in turn is con-nected to foie gras, while having a negative atti-tude towards the latter and a favourable preference for the first two. What diffuses via social networks are the pairwise links between these elements, and these links are strengthened (reinforced) the more people are exposed to these associations (e.g., caviar and champagne) when interacting with others (alternatively, associations that are not reinforced in interaction decay over time). Each time a person is exposed to a given cultural asso-ciation, the strength between the two is 'updated' in the cultural network, meaning people harmo-nise their attitudes towards the things they think are related, much like people need to 'balance' their sentiments across triadic structures in inter-personal sentiment networks (Davis, 1963). Thus, the person who thinks caviar is strongly related to both champagne and foie gras but who hates foie gras must either start liking foie gras more or lik-ing champagne and caviar less. The ADM thus links the idea of schemas as a network of cultural elements to interaction, social construction and learning processes and imple-ments this dynamic in an elegant computer simulation. One pay-off is that the ADM can show that the core phenomenon that those who study dif-fusion believe requires structural segregation at the network level (attitude or taste polarisation) can emerge via associative diffusion mechanisms even when social structures are not segregated (e.g., fully connected or high-density networks), a situation where ‘virus-type’ diffusion models – such as Mark’s network ecology model – would predict convergence rather than cultural niche divergence. This is a significant result because it attests to the robustness of the cultural polarisation phenomenon even in the face of structural condi-tions that network ecology models would predict would make it impossible, while also showing that a critical boundary condition of these models – pre-existing relational segregation – is not neces-sary for the emergence of cultural differentiation. The ADM also complicates ‘network-based’ solu-tions to the polarisation issue premised on creating links or increasing exposure between people with opposed attitudes and worldviews, as suggested in recent empirical work (Bail et al., 2018) the cultural holes Imagery As DiMaggio (2011) noted more than a decade ago, research on ‘cultural networks’ has become a central organising hub linking work on creativity, collaboration, group identity, symbolic boundaries Culture and networks 193 **and the relational constitution of meanings in the sociology of culture, and network imagery is a natural way to unify this otherwise disparate array of work on a variety of substantive areas. Mützel & Breiger (2020) note that much of this work was united by the duality between two or more entities in (multimode) cultural networks. Nevertheless, while hinting at an underlying theoretical model, the overall idea of cultural networks and the more specific idea of duality are more like general frameworks helpful in organising a variety of empirical work around loosely related thematic areas. Is there an underlying theoretical model that can be seen as doing the bulk of the explanatory work? I propose that there is such a model centred on the notion of cultural holes (Pachucki & Breiger, 2010).** Cultural Holes as a Model for Cultural Networks In contrast to the constructural (or associative dif-fusion) model, there has never been a formalisa-tion of the cultural holes model.2 However, surveying the use of the idea in recent work is not difficult. The cultural holes argument can be seen as a steady-state macro-level implication of the constructural model but applicable to social sys-tems at scale. In this case, differentiation and insulation generate discontinuities in the cultural distribution of knowledge across people (Reay, 2010), thus creating ‘gaps’ in the cultural struc-ture, represented by ‘patterned absences of rela-tions’ among cultural items (Breiger, 2010, p. 39). This contrasts with the small group, where the steady-state equilibrium of the constructural pro-cess is for everyone to end up knowing everything everyone else knows. Second, as Carley (1991) noted, there is a link between this imagery and that of duality since the cultural distribution of knowledge can be modelled as a two-mode net-work of people by ‘pieces’ of cultural information. Projecting the network into a person-by-person one-mode weighted network in the way proposed by Breiger (1974), we end up with the cultural similarity between people, as discussed earlier. Conversely, projecting the original two-mode network into an item-by-item one-mode network gives us the cultural relatedness of the items, such that two cultural items are strongly related if they are likely to be known by the same people and weakly related if they are unlikely to be known by the same people. We can thus define a ‘cultural ego network’ at the agent level, with the person connected to all the cultural items they know – the link between the items being the weight of their linkage in their respective one-mode network pro-jection, perhaps suitably corrected for the statis-tical likelihood of observing a weighted link of a specific size (Neal, 2014). Computing Burt’s ([1992]2009) ego-network efficiency among the alters (cultural items) of the cultural ego network gives us one way of measuring the extent to which agents bridge cultural holes (Lizardo, 2014). In essence, the cultural holes argument links the cultural distribution of pieces of cultural knowledge in a given social system to define the ‘position’ of a given agent in the ‘sociocultural’ structure (Lee & Martin, 2018). Most research using this approach links the positional status of agents or items to various outcomes, much like previous work on sociometric holes (Burt, 2004). Note, however, that this ‘agent-centric’ approach to studying cultural holes, although natural and intuitive, is not the only one we can take. Given duality, it is possible to be interested in the item-by-item ‘culturosocial structure’ itself (Lee & Martin, 2018) and study cultural holes from a ‘culture-centric’ perspective – for instance, by looking at which cultural items are likely to be ‘bridges’ (e.g., mediating between clusters of weakly connected items). In this respect, the cultural holes argument presupposes a ‘mutual alignment’ between agent and culture-centric perspectives (Puetz, 2017). Accordingly, analysts have taken both agent- and culture-centric routes in analysing cultural holes. Agent-Centric Approaches to the Study of Cultural Holes **Cultural Holes, Omnivorousness and taste A natural application of the cultural holes argu-ment from an agent-centric perspective is to specify phenomena that have been recalcitrant to analytic treatment in other fields. One particularly promising arena is the sociological study of cultural taste and consumption (Bourdieu, [1979]1984). This area has come to be populated by a host of concepts meant to denote ‘openness’ and ‘cosmopolitanism’ as a modern marker of being a ‘tolerant’ person with a taste for diversity and varieties of cultural experiences (Ollivier, 2008). These include, in addition to the epony-mous ‘cosmopolitanism’, such constructs as ‘cul-tural omnivorousness’ (Peterson & Kern, 1996). For example, suppose we view cultural taste as ‘pieces’ of cultural knowledge. In that case, the natural thing is to see the cosmopolitan ‘cultural omnivore’ as one who bridges cultural holes via their consumption choices.** The SAGe hAndbook of SociAl neTwork AnAlySiS194 Lizardo (2014) follows this approach, not-ing that the cultural omnivore can be considered an agent with an 'efficient' cultural ego network relative to cultural tastes. That is, omnivores select non-redundant genres, with redundancy defined as the extent to which the audience of any one genre chosen overlaps with that of other gen-res also chosen – building on the logic of Burt's structural holes argument. From this perspective, omnivorousness refers to bridging cultural holes in a space defined by cultural genres by choos-ing sets of genres whose audiences display rela-tively low overlap. After proposing an intuitive way to capture this tendency, combining Breiger's (1974) projection approach with a variation of Burt's metrics for structural holes, Lizardo shows that the socio-demographic markers traditionally associated with cultural omnivorousness when measured as the sheer quantity of choices differ in systematic ways from those linked to omnivorous-ness when measured in terms of bridging cultural holes. In more recent work, Puetz (2021) extends the cultural holes argument concerning aesthetic engagement to the realm of friendship prefer-ences, showing that people who bridge cultural holes when it comes to their musical taste choices are also more likely to express a preference for friends who are 'creative' and 'cultured' as com-pared to people who make cultural choices that commonly go together. Cultural Holes and the Categorical Imperative Silver et al. (2022) deploy the cultural holes argu-ment to shed light on the 'categorical imperative', namely the often-noted phenomenon that actors (both individual and corporate) who cross categori-cal boundaries are penalised in market settings. Like the above, this work shows that recent work on 'categories' in organisation and management stud-ies (Goldberg et al., 2016a; Kovács & Hannan, 2015) can most profitably be specified with the cultural holes framework. The basic idea is that when audiences (regulators, critics, customers) are faced with actors who present themselves in multi-ple categories (e.g., a brewery that is also an ice creamery), they are confused and infer that a 'jack of all trades is probably master of none', or more accurately not as masterful as one with a focused (e.g., singular) identity (Hannan, 2010). From this perspective, bridging cultural holes in a category system is likely to be deleterious to actors in com-petitive fields. By contrast, Silver et al. (2022) argue that rather than an unconditional negative link, the connection between cultural-hole-spanning, crea-tivity, and success is likely to take something closer to an inverted u-shape, especially in fields that reward innovation and creativity. Too much focus on a single genre category (lack of cultural-hole bridg-ing) is seen by audiences hungry for novelty as con-ventional and boring. However, spanning cultural holes across considerable categorical distances is likely to trigger the various deleterious mechanisms mentioned earlier, with audiences left befuddled at bizarre or non-standard genre combinations. **To test their proposal, Silver et al. examined data from close to 3 million bands listed in Myspace in the late aughts. They use the music genre catego-ries selected by each band and compute the cat-egorical distance between genre labels based on co-**

occurrence patterns from the one-mode (genre-by-genre) projection of the original band-by-genre two-mode network. They find that indeed, across three musical ‘worlds’ – high-level clusters of genres labelled ‘Rock’, ‘Hip Hop’ and ‘Niche’ (see also Silver et al., 2016) – there is an inverted u-relationship between cultural-hole bridging and popularity (as measured by the number of views, fans and digital ‘plays’ of their music). Bands with highly focused identities (failing to bridge cultural holes) and bands who span cultural holes across very distant categories are less popular than bands who combine categories at a moderate distance from one another. Thus, moderate cultural-hole bridging is the key to success in this empirical case. While this is the predominant tendency in the data, Silver et al. also find systematic varia-tion in this empirical pattern by musical world.3 In some musical worlds, like Hip Hop, the predomi-nant pattern is closer to a ‘dual innovation’ model, with peaks in popularity at both a moderate and an extreme label of cultural-hole bridging. Some musical worlds reward creativity and unconven-tionality, but most follow the tendency to reward cultural-hole bridging up to a point, after which the ‘categorical imperative’ kicks in.4 Cultural Holes versus Structural Holes Silver et al.’s work proposes a systematic link between cultural-hole bridging, creativity and success; Choi’s (2018) recent work examines this issue directly. Choi uses the cultural holes argu-ment to study the origins of ideational creativity among members of the IT department of a medium-sized tech company in South Korea. The topic of ‘good ideas’ has been studied before using sociometric network models like Burt’s structural holes (Burt, 2004). The primary finding here is that people who bridge sociometric struc-tural holes are likelier to have ‘good ideas’ than those who live in social worlds in which their contacts are connected. The presumed mechanism is that structural holes provide the potential to Culture and networks 195 access otherwise disconnected ideas by others. Thus, structural brokers are better positioned to synthesise these separate pieces of information into more original and creative insights. **Choi (2018) reasons that the structural holes model is an indirect ‘one step removed’ version of a more basic cultural holes model. That is, structural brokers have better ideas because they are cultural brokers; the direct operative mechanism concerns not the position of the person in a sociometric net-work of personal relations but a people’s position in a cultural network composed of local ideational frames. Regarding creativity, the cultural holes argument partially supersedes the structural holes one (or, at the very least, provides a more direct mechanism leading to the same outcome). To test this hypothesis, Choi first measured the most prev-alent local frames in the company using a free-text elicitation strategy from which higher-level frames were coded and extracted, then used an ‘idea tour-nament’ approach based on**

Salganik’s Wiki survey method (Salganik & Levy, 2015).5 In this setup, participants anonymously submit-ted ideas to improve the company. Then, the ideas were (also anonymously) rated by others using pairwise comparisons between two randomly selected ideas at a time. Ideas that were more likely to ‘win’ when compared to other randomly selected ideas were deemed the most creative. The frame extraction approach resulted in a two-mode network of people by cultural frames. Cultural brokers are thus people who hold cultural frames not likely to be held by the same others. Choi found that, indeed, cultural broker are more likely to have ideas considered creative by others, even after adjusting for sociometric brokerage (which also leads to ideas being considered more creative) and relative cultural fit (the likelihood of holding frames that are also likely to be held by others). **Choi’s finding that bridging cultural holes has a direct effect on creativity even after statistically adjusting for bridging sociometric structural holes is consistent with recent work arguing and showing that sociometric bridging (structural holes) is ana-lytically distinct from cultural bridging (cultural holes) and thus has independent effects on key out-comes. For instance, Graham et al. (2022) analyse large-scale scientific co-authorship data across 25 scientific fields. They use computational linguistic techniques to measure each scientist’s diversity of information based on the terms and concepts used in their publication’s title, abstract and keywords and the articles they cite in their work. They find only weak correlations between sociometric posi-tion in the co-authorship network and ego’s direct and indirect access to information diversity, show-ing that bridging structural holes does not automat-ically translate into bridging cultural holes. Graham et al.’s findings dovetail with those of Goldberg et al. (2016b), who also use computa-tional linguistic techniques to measure the extent to which people exhibit ‘culturally fit’ (e.g., use language that is similar to their communication partners in a directed email network) within an organisation. They find that bridging structural holes and having a high cultural fit lead to more beneficial outcomes (e.g., better performance reviews and lower chances of involuntary exit) when considered independently. However, pre-cisely because the link between position in the cul-tural and sociometric networks is relatively weak, the two dimensions can also be considered jointly (e.g., one effect moderating the other). When this is done, the results indicate that structural holes help but only for those who culturally fit the organisation (‘assimilated brokers’). Notably, the results suggest that bridging cultural holes is suf-ficient for experiencing comparatively successful outcomes. People with high levels of sociometric constraint but who stand out culturally from others in the organisation – integrated non-conformists – do systematically better compared to those who have access to structural holes but do not**

bridge cultural holes ('disembedded actors'). Finally, work by de Vaan et al. (2015) shows that the argument distinguishing cultural from structural bridging extends beyond individuals to explain the creative success of teams. While previ-ous work on creative teams (e.g., Uzzi & Spiro, 2005) focuses mainly on structural network posi-tion – and is thus forced to infer cultural bridging from structural bridging – de Vaan et al. develop separate measures of each for creative teams of video game developers. Furthermore, they show that it is a combination of structural and cultural bridging that best explains creative success – as measured by critical acclaim teams composed of groups (defined as subsets of creators who have worked together in the past) that share members with other groups within the team ('structural folding') and groups that bridge cultural holes by spanning large distances in categorical product space relative to other groups (as defined by the distance in product space of previously created games created) are the most likely to come up with both novel (category-atypical compared to previ-ous entries) and acclaimed cultural products. Culture-Centric Approaches to the Study of Cultural Holes Cultural Consumption Networks Sokolova and Sokolov (2020) set out to study cultural holes by building a culture-centric The SAGe hAndbook of SociAl neTwork AnAlySiS196 **network of literary authors based on patterns of fiction books co-borrowing from the municipal library network in St Petersburg, Russia. Authors are strongly connected to the extent that there is significant overlap between the sets of people who read them. Sokolova and Sokolov's primary goal is to ascertain whether institutionally consecrated objects (or objects consumed by high-status people) – which in Lizardo's original definition would count as 'asset-specific' or 'strong' culture in Schultz & Breiger's (2010) sense – can them-selves bridge cultural holes in a culture-centric network. They find, contrary to the idea that insti-tutional "consecration" (in Bourdieu's sense) ren-ders high-status cultural goods 'niche', that the most consecrated authors are also the ones most likely to bridge cultural holes – as given by Burt's constraint measure and betweenness centrality – and the same goes for authors more likely to be read by college-educated audiences. These results are consistent with Lewis & Kaufman's (2018) warnings regarding the relativity of exogenous markers of the status of cultural goods, as the latter have no logical link to whether they serve as strong or weak culture since these last speak to formal properties of such goods within a local ecology. Sokolova and Sokolov's study shows that in a context where consecrated and high-status works are generally valued, they behave very much like we would expect 'popular culture' would; as bridges within the cultural network. Cultural Holes in Text Networks Perhaps the most exciting recent advance in the culture-centric analysis of**

cultural holes is Stoltz & Taylor's (2019) proposal to measure 'discursive holes' in-text similarity networks. The recent rise of computational social science has brought a slew of techniques to analyse large-scale text data. However, almost all the now well-established methods yield some weighted text-to-text network where the edge weight is based on the similarities – and, by implication, the distances – between each pair of texts. Although each analytic tech-nique – for example, topic models, word-vector embeddings – produces a different criterion for similarity, every text analyst deals with such a network. For instance, Stoltz & Taylor note that topic models yield a two-mode matrix of texts by topic, where the row entry for text is the probabil-ity that text i engages topic j (with the probabili-ties summing to one row-wise). This argument can be generalised to the analysis of two-mode network data more generally beyond texts since every one-mode projection is, in fact, a potential similarity (or distance) matrix across the nodes in each mode (Everett & Borgatti, 2013). Following this insight, Stoltz & Taylor use a measurement strategy developed for agent-centric cultural anal-ysis but deploy it in the culture-centric level of texts in discursive fields. They thus propose a measure of 'textual spanning' (cultural-hole bridging) for particular texts based on a simple idea: within a discursive field defined by a textual similarity space, texts bridge cultural (discursive) holes when they are similar to texts that are them-selves dissimilar. Stoltz and Taylor go on to show that this approach to measure cultural-hole span-ning generates insights about the positionality of texts in discursive space that are occluded by using standard 'medial' or 'radial' centrality measures based on the weighted path distance (like betweenness and closeness). Cultural Holes, Atypicality and Categorical Diversity While not couched in explicit network terms, Goldberg et al.'s (2016a) approach to analysing cultural boundaries is based on the fundamental cultural holes imagery. Importantly, their approach can be interpreted as engaging in agent- and culture-centric analysis simultaneously by defin-ing agent-centric properties relevant to cultural-hole bridging using culture-centric ones.6 The critical difference is that rather than departing from a two-mode network of people by cultural objects chosen, they develop concepts and metrics linked to a three-mode network of people, objects and the category labels applied to those objects (e.g., by audiences like critics, consumers, or web-sites). The key observation from the perspective of the cultural holes model is that sets of objects in a given cultural domain (e.g., movies, cuisines, art and the like) and sets of labels (Action, Cambodian, Cubist and the like) can bridge cultural holes by spanning wide distances in label space. People, in turn, can bridge cultural holes by choosing cate-

gorically distinct objects or objects that bridge across category labels. For instance, we can begin by constructing a two-mode object-by-label network – where object i is linked to label j if that label is applied to that object (Kovács & Hannan, 2015). We can then use variations of the Breiger (1974) projection approach to define a one-mode label-by-label net-work (Everett & Borgatti, 2013). In this network, strongly related labels have significant overlaps between the set of objects each label is applied. On the other hand, category labels are weakly related when the overlap between the label sets is relatively small (Silver et al., 2022). In this way, the weighted label-to-label network encodes the relative similarity of each pair of labels. By implication, this network also encodes inter-label Culture and networks 197 distance since, as Hannan and Kovács (2015) note, similarities between labels are an inverse function of distance in some conceptual space (Gardenfors, 2014). Strongly related labels are close, while weakly related labels are far. Accordingly, we can construct a derivative label-to-label network where the weighted links are given by pairwise distances, using some suitable mathematical transformation of the original pairwise similarities. **Given that each object is assigned a set of labels, we can then aggregate or average the pairwise distances of the labels assigned to each object. According to this metric, atypical objects will receive high scores, while typical objects will receive lower scores. This is another way of saying that atypical objects bridge cultural holes across labels. We can also use the inter-label distance information to define an object-by-object network. The weighted link between objects is given by the average similarity between the two sets of labels assigned to the objects incident to each edge. The inter-object network thus encodes the relative cate-gorical distinctiveness between each pair of objects, with categorically non-distinct objects clustering together and categorically distinct objects falling in separate communities. As with labels, we can generate a derivative inter-object network by trans-forming inter-object similarities into distances. With this information in hand, we can loop back to the agent's perspective and define a person's taste for atypicality by aggregating or averaging the atypicality scores of the objects they choose (as encoded in a two-mode network of people by objects chosen).7 Thus, agents can bridge cultural holes by selecting atypical objects, namely objects bridging culture holes in label space.8 However, there is a second way to define an agent's penchant for bridging cultural holes in their consumption choices. Similarly, we can define a person's taste for variety by aggregating or averaging the pair-wise distances of the objects they choose. Thus, agents can bridge cultural holes by selecting cat-egorically distinct (sets of) objects. As Goldberg et al. (2016a) show, these two agent-centric ways of bridging cultural**

holes are conceptually and empirically distinct, as the atypicality of an object relative to the set of labels it is assigned is not defi-nitionally linked to its categorical distinctiveness relative to other objects. Importantly, tastes for variety and atypicality are empirically distinct from the sheer numeri-cal quantity of objects chosen – what has been referred to in the literature as 'omnivorousness by volume' (Lizardo, 2014). In the case of atypical-ity, for instance, a person can bridge cultural holes even when selecting a single (e.g., highly atypi-cal) object – 'univores' are therefore capable of bridging cultural holes. Alternatively, a person can select many categorically non-distinct objects, thus failing to bridge in the variety sense – quan-titative 'omnivores' are therefore not guaranteed to bridge cultural holes. Because they are distinct, agents may have a high or low taste for atypical-ity combined with a high or low taste for variety, yielding three ways to bridge cultural holes. For instance, people can choose (or like) categorically a small set of indistinct atypical objects ('mono-mixer'), a large set of categorically distinct typical objects ('poly-purist'), or a large set of categori-cally distinct atypical objects ('poly-mixer'). **While being a poly-mixer may seem like the ulti-mate way to bridge cultural holes, Goldberg et al. (2016a) provide suggestive evidence that the 'high-status' omnivore who populates much work in the sociology of taste (see Jarness & Friedman, 2017) is a person who bridges cultural holes in the variety sense, but who refuses to do so in the atypicality sense. These poly-purist omnivores decide instead to 'police' categorical boundaries (e.g., by down-grading atypical objects), thus sharpening the typi-cality of objects and increasing opportunities for cultural-hole spanning in their preferred sense.** concludIng remarks **We opened the chapter by considering whether 'culture and networks' is a research speciality, a research cluster, or something else. I noted that the answer to these questions depends on the degree of unity we seek. On the one hand, we can be content with a rather superficial thematic unity based on topics and substantive emphases. On the other hand, we can seek a deeper unity based on the underlying theoretical models – or, more loosely, as referred to here, 'imageries' – analysts rely on to develop theories, formalisation, meas-ures and hypotheses in their work. I argue that such underlying unity does exist in the field and is centred around two underlying ur-models: con-structuralism and cultural holes. I showed how research using this imagery has progressed by addressing a coherent set of problems and issues and by extending theoretical models and claims to speak to the limitations of previous formulations.** Constructuralism For instance, regarding constructuralism, Mark built his network ecology model on partial con-structuralist insights, and Lizardo's conversion model developed the constructuralist strand left behind in the network ecology formulation. The SAGe hAndbook of SociAl neTwork AnAlySiS198

Subsequently, Lewis and Kaufman responded to weaknesses in the conversion approach, generalis-ing and distinguishing between various conver-sion mechanisms while linking to the larger cultural ecology. Finally, Goldberg and Stein's associative diffusion model develops the some-what rudimentary cognitive foundations of origi-nal constructuralism by incorporating insights from the schematic construction of cultural mean-ing while incorporating cognitive consistency mechanisms into the interactionist constructionist imagery. The ADM thus moves beyond the limita-tions of network ecology, constructuralism and conversion models, all of which are tied to limit-ing conceptions of piecemeal cultural diffusion. The cutting edge of future work in this area thus lies in further integration between social network, interactionist, cultural and cognitive mechanisms in processes of cultural creation, transmission and diffusion and in generating observable pat-terns of cultural niche creation, maintenance and dissolution – for example, in the vein of Fuhse (2021), Goldberg & Stein (2018) and de Vaan et al. (2015). Importantly, this will require more sophisticated models of how culture is internal-ised and processed by people embedded in social network structures and subject to interactional constraints and asymmetries (Arseniev-Koehler & Foster, 2020; Foster, 2018; Lizardo, 2021). Additionally, we will have to keep firmly in sight the double emphasis of constructuralism as both a theory of what networks do and a theory of where networks come from.9 Cultural Holes Research on cultural holes has also developed rapidly since Pachucki & Breiger (2010) intro-duced the idea, with accompanying innovations in measures, formalisation and the range of substan-tive applications. I argued here, however, that behind the thematic unity of the various studies, and even above and beyond the underlying con-ception of 'duality', there is an underlying theo-retical imagery, linking agents and cultural objects (broadly defined) to mutually defining positions in a field (Puetz, 2017). As we saw, the cultural holes models can even be seen as a positional, comparative-statics-cousin to constructuralism's cognitive interactionism since both models share a common strand in Breiger's (1974) duality idea. Accordingly, the cultural holes argument relates the agent's and object's position in socio-cultural and cultural-social structures to various substantive outcomes (e.g., creativity, spanning, cosmopolitanism, vision) of interest to the analyst. In this particular way, the cultural holes argument has been interpreted and used similarly to its sociometric 'structural holes' cousin. As we saw, however, a key implication of recent work in this area is that cultural and structural hole bridging must be kept analytically distinct; one cannot be inferred from the other as has been the practice of assuming that structural brokers are also cultural brokers (Goldberg et al., 2016b; Graham et al., 2022). The cultural holes model is distinct from its sociometric cousin because it conceives the 'position' people occupy in sociocultural space as driven by analytically distinct (but coupled) dynamics from those driving the construction and the dissolution of sociometric ties. Recent advances in measuring culture from free-form text data using computational linguistic techniques have thus helped clarify the empirical contribution that people's position in the cultural information space has on key outcomes, as distinct from their 'structural' position in a network of social ties. **Another advantage of treating the idea of cul-tural holes as an underlying ur-model is that we can then integrate work that, while not nominally claiming to belong to this camp, formally does so, like work on categories, cognition and category-spanning in organisation and management studies (Goldberget al., 2016a; Hannan et al., 2019; Silver et al., 2022); this is something that would have been impossible had we remained at the more surface level of commonalities based on explicit concepts or formal modelling procedures. Nevertheless, as work using the cultural holes imagery moves for-ward, more explicit formalisation of the cultural holes argument, perhaps a deepening of the idea of cultural holes in a more dynamic vein (e.g., towards temporal two-mode networks), such as by more explicit use of dynamic computational models and simulation – à la constructuralism and associative diffusion – seems to be called for, since most work in the cultural holes approach uses static measures of position. In the same way, an extension of the cultural holes imagery outside fields where it has been exploited the most (e.g., cultural production and consumption), such as knowledge production in science (Vilhena et al., 2014), or entrepreneurship and innovation – as with Choi's study mentioned earlier – would help stretch the paradigm and lead to further concep-tual and empirical development.** Notes 1 The field has also been covered in a monograph-length publication by McLean (2016). 2 Of course, Breiger (1974) formalised the idea of duality in his classic paper, which is central to the cultural holes argument. Culture and networks 199 3 See unconventionality.github.io/ 4 Socio-demographic and political-economic char-acteristics of the metro area also moderate the strength of the inverted-u relationship between cultural-hole bridging and success without violat-ing the general pattern. 5 See www.allourideas.org/ 6 See Kovàcs (2010) and Lizardo (2018) for related approaches. 7 Goldberg et al. (2016a) distinguish 'sampling' an object from 'liking' an object (technically defin-ing a two-mode multigraph with 'sampling' and 'liking' edges between people and objects), but that is a complication that is not relevant to our purposes. 8 In the same way, community detection on the inter-object network yields clusters of categori-cally similar objects separated from other sets of categorically dissimilar ones. 9 That is, constructuralism functions as both a 'net-work theory' and a 'network theory of networks' in Borgatti and Halgin's (2011) sense. references Arseniev-Koehler, A., & Foster, J.G. (2020). 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